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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

IV B.TECH I SEM | MAJOR PROJECT DEVELOPMENT & THESIS

AI CodeSense

Intelligent Code Review and Documentation Assistant

Batch Number : Batch 6

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Overview of the Presentation:-

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Abstract :

➢ AI CodeSense is an AI-powered web platform that supports developers with automated code review, bug detection, and technical documentation.

➢ Users can upload source code, paste code snippets, or connect GitHub repositories for automated analysis.

➢ Artificial Intelligence detects bugs, identifies code smells, analyzes code complexity, and recommends performance improvements.

➢ The platform generates technical documentation, UML descriptions, and offers an AI-powered chatbot for intelligent code explanations.

➢ Built with React.js, Spring Boot, MySQL, Python (FastAPI), and the Google Gemini AI API for a scalable, unified developer experience.

Objective :

- Automate code review to detect bugs, code smells, and quality issues using AI.
- Analyze code complexity and provide actionable performance improvement recommendations.
- Automatically generate technical documentation and UML descriptions from source code.
- Provide an AI-powered chatbot for real-time code explanations and developer assistance.
- Support code upload, snippet paste, and GitHub repository integration for seamless analysis.
- Improve development efficiency, code quality, and documentation consistency across projects.

Introduction

➢ Software development requires continuous code review, bug detection, security analysis, and technical documentation to ensure quality and reliability.

➢ Developers often rely on multiple independent tools for analysis, security scanning, and documentation, resulting in a fragmented workflow.

➢ AI CodeSense envisions a single, intelligent platform that unifies these tasks using Artificial Intelligence.

➢ The platform is built using React.js, Spring Boot, Python (FastAPI), MySQL, and the Google Gemini AI API.

Existing System :

- ➢ Manual code reviews are time-consuming and prone to human oversight, especially in large codebases.
- ➢ Developers depend on multiple disconnected tools for code analysis, security scanning, and documentation generation.
- ➢ Technical documentation is frequently incomplete, inconsistent, or quickly becomes outdated.
- ➢ No single existing platform combines bug detection, code quality analysis, and AI-assisted documentation.
- ➢ This fragmented workflow reduces development efficiency and increases the likelihood of overlooked issues.

Proposed System :

➢ **AI CodeSense:** an AI-powered code review and technical documentation platform for the full software development lifecycle.

➢ Users upload source code, paste snippets, or connect GitHub repositories for automated analysis.

➢ The Google Gemini AI API detects bugs, identifies code smells, and analyzes code complexity.

➢ The system recommends performance improvements and automatically generates technical documentation.

➢ Creates UML descriptions and provides an AI-powered chatbot for intelligent, conversational code explanations.

➢ Delivers a scalable solution that replaces fragmented manual workflows with one unified platform.

Modules :

- **User Authentication Module:**
  Registration, login, password management, and role-based access control for developers and admins.
- **Code Input Module:**
  Supports file upload and GitHub repository connection for automated analysis.
- **AI Code Analysis Engine:**
  Detects bugs and code smells, and analyzes code complexity via the Python (FastAPI) service and Gemini API.
- **Documentation Generation Module:**
  Automatically generates technical documentation and UML descriptions from analyzed code.
- **AI Chatbot Module:**
  Conversational assistant that explains code, answers developer queries, and suggests fixes.
- **Reporting & Dashboard Module:**
  Displays analysis results, code quality scores, and historical review reports.

System Requirements :

SOFTWARE REQUIREMENTS

- **Operating System:** Windows / Linux
- **Text Editor:** VS Code, Eclipse, IntelliJ IDEA
- **Frontend:** React.js, HTML, CSS, JavaScript
- **Backend:** Spring Boot (Java), Python (FastAPI)
- **Database:** MySQL/MongoDB
- **AI Engine:** Google Gemini AI API

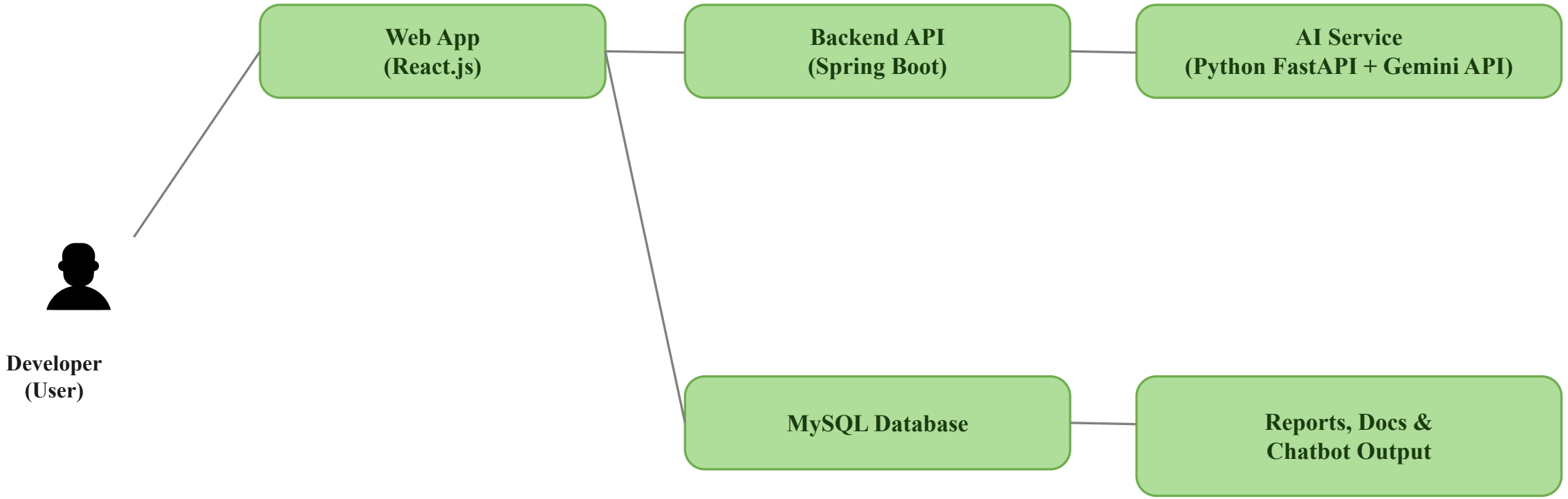
HARDWARE REQUIREMENTS

- **Processor:** Intel Core i5 or higher
- **RAM:** 8GB or higher
- **Internet:** Broadband connection, minimum 10Mbps

Algorithm :

- **Static Code Analysis Algorithm :** Parses source code (AST-based) to detect syntax issues, bugs, and unsafe patterns.
- **Code Complexity Algorithm :** Computes cyclomatic complexity and maintainability index for each function or module.
- **Code Smell Detection Algorithm :** Applies rule and pattern-based checks to flag anti-patterns and quality issues.
- **AI Documentation Generation Algorithm :** Uses the Gemini LLM to summarize code logic into technical documentation and UML descriptions.
- **Chatbot Response Algorithm :** Combines contextual code retrieval with the Gemini API to answer developer queries.

System Architecture :



The developer interacts through the React.js web app, which routes requests through the Spring Boot backend. The Python (FastAPI) AI service calls the Google Gemini API to analyze code, while MySQL stores user, project, and report data. Results are returned as reports, generated documentation, and chatbot responses.

Results :

- Login & Registration screens for secure developer access.
- Code Upload / GitHub Connect dashboard for submitting projects for analysis.
- AI Analysis Report view showing detected bugs, code smells, and complexity scores.
- Auto-generated Documentation & UML description viewer.
- AI Chatbot interface for interactive code explanations.

Screens will be captured and added here once the corresponding modules are implemented.

Conclusion :

➢ AI CodeSense integrates automated code analysis, technical documentation generation, and AI-assisted developer support into one scalable platform.

➢ The system reduces manual review effort, improves code quality, and keeps documentation consistent and up to date.

➢ The solution lays a strong foundation for building more advanced, AI-driven developer productivity tools.

Future Scope :

➢ CI/CD & Language Support

- Integrate with CI/CD pipelines to trigger automated reviews on every commit or pull request.
- Support additional programming languages and deeper security vulnerability scanning.

➢ Collaboration & Intelligence

- Add team collaboration features such as inline review comments and analytics dashboards.
- Enhance the AI chatbot with voice interaction and direct IDE plugin integration.
- Introduce personalized recommendations based on a team's historical coding patterns.

Any Queries?

Thank You